

Planning with justice: Using spatial modelling to incorporate justice in electricity pricing – The case of Tanzania

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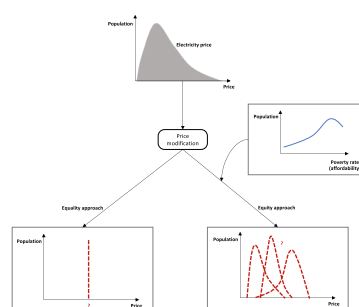
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HIGHLIGHTS

- A methodology to incorporate *just* pricing in Tanzania's electrification planning.
- *Equality* pricing means a constant price throughout, considering only technology cost.
- *Equity* pricing takes into account the spatial distribution of poverty in the country.
- Grid customers bear the burden of price increase in *both* justice scenarios.
- Based on prices, capex and tariff subsidies values are calculated.

GRAPHICAL ABSTRACT



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ABSTRACT

Universal electrification by 2030 is an important goal of Sustainable Development Goal (SDG) 7. Electricity provision no longer relies only on centralized grid expansion, but also on off-grid and mini-grid systems. Although this technological diversity holds promise, the technologies differ both physically and institutionally in electricity delivery. These differences raise equity and justice concerns around how they are implemented. For example, how can electricity be kept affordable for all consumers when access is provided by various technologies operated under different business models? This paper addresses this aspect of affordability and sheds light on how the SDG7 target could be met more equitably and fairly. We use a novel analytical methodology to apply two different principles of justice – equality and equity – to incorporate affordability into electricity pricing. Using a geospatial electrification model and Tanzania as a case study, we first arrive at price levels based on the principle(s) of justice. Then, we produce location-specific recommendations for subsidy levels needed to ensure those price levels. We find that the equity approach benefits a bigger section of the population than the equality approach. Moreover, the former costs significantly less per capita than the latter. Having said that, the equity approach is complex and therefore harder to implement. The methodological framework proposed in this study acts as a proof-of-concept for examining concerns around distributive justice using quantitative energy modelling tools and drawing policy relevant insights for energy planning in developing countries. Additionally, by

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focusing on the spatial aspects of energy access and the issue of fairness, the study also contributes to the growing conceptualizations of energy justice.

1. Introduction

Globally, ~840 million people lack access to any form of electricity [1], and ambitions for universal electricity access are enshrined in the Sustainable Development Goals (SDGs) as SDG 7. The goal aims for ensuring access to *affordable, reliable, and sustainable* energy for all. Upholding all three principles here could be challenging. As far as sustainability is concerned, renewable energy (RE) based distributed generation (DG) systems have been a boon for energy access. They have not only diversified the technological solutions, but also done so utilizing clean resources. The number of people benefitting from off-grid renewable energy solutions – both mini-grids and standalone systems – have grown six fold between 2011 and 2016 to 133 million [2]. They are expected to provide almost 60% of new access till 2030. These technologies, however, are different compared to the incumbent (centralized grid) technology. They are smaller in scale and rely on spatially diverse resources, which makes energy provision and planning more complex. This complexity has implications for the other two aspects of the SDG7 goal – *affordability, and reliability*, and this study explores the first of those. Through a quantitative modeling approach, we examine the following research question: *Given a technologically and spatially diverse solution pool, how could electricity provision be made affordable for everybody in a specific geography?*

In the context of meeting SDG7 using RE based DG, the cost of electricity provision varies spatially, as well as across technologies [3]. Extension of the centralized grid to remote rural areas is investment heavy and expensive mainly because of low demand density and remoteness [4]. Such conditions make RE based DG cost competitive in rural electrification [4], although as compared to urban (grid) electricity they may still be more expensive. Moreover, on the consumer side, the ability to pay (affordability) varies depending on the socio-economic conditions. Affordability of remote and rural consumers for example, could be lower than their urban counterparts [5]. Ignoring such indicators in electricity provision raises the issue of fairness and *justice* from the consumers' standpoint. For centralized grids, countries have relied upon tariff regulations to make electricity affordable for end consumers – and thus address concerns around equity and fairness. However, the same kind of policy fix is difficult to implement for DG technologies, especially when they are not grid connected. Thus, in the absence of any appropriate regulatory mechanism, RE based DG electricity may remain unaffordable for many remote and poor customers even if it is the least cost option for that area. By exploring questions around *fairness* through a spatial analysis, this study produces valuable policy-relevant insights for facilitating technologically diverse electricity provision.

The research question above specifically addresses two concerns: a) facilitating economical electrification via diverse technology options and b) fostering *fairness* in the consumer prices around these technologies. The objectives of the study are thus two-fold:

1. To develop and demonstrate a methodology based on an electrification modelling tool to explore questions around justice/fairness in electricity pricing (for developing countries)
2. To understand the implications of two scenarios:
 - a. Everybody pays the same price for electricity (*equality* scenario)
 - b. Prices vary spatially with respect to affordability (*equity* scenario)

With the above objectives, we incorporate *fairness* into electricity prices in two different ways. One way replaces the price distribution with a constant average price (equality), and the other modifies the

distribution to incorporate affordability (equity). The modified prices are set such that the overall revenues from electricity sales remains constant, i.e. one part of the population subsidizes the other part. We use poverty rate as a proxy parameter for incorporating affordability in a given location. We therefore proceed to calculate the two types of subsidy distributions – tariff subsidy for the population getting grid electricity and capex subsidy for the population getting electricity from RE based DG (standalone or mini-grid) systems. Before we describe the methodology in specific detail however, we elaborate on the research context of this study below.

1.1. Literature review

This study mainly touches upon two important areas of energy research. First is the use of 'systems modeling' approaches to solve energy issues and the other is the growing field of 'energy justice'. Within this, we also take a deeper dive into the notion of *justice* itself, in order to better understand the analytical framing used here.

Within the energy sector, planning generally entails the use of energy system models. These models deploy different computing methods to provide valuable insights to guide energy related decisions. There are myriad energy system modelling tools used by developing countries to make economical energy decisions through long-term energy planning, [6]. Models are often used to identify the optimal operational strategy for generating heat and/or electricity for a given energy system [7]. The energy modelling landscape varies from economy-wide 'top-down' models to disaggregated 'bottom-up' models. These models have a crucial role in energy planning with the objective of taking the most cost-effective investment decisions under various constraints: resource, socio-economic, and environmental. Given the spatially distributed nature of alternative energy resources, and remoteness of rural communities, GIS based modeling tools have been a significant recent development in the analysis for energy access solutions [8]. For any specific location, the knowledge of population density, fuel costs, distance from the central grid, demand level and income can help determine site-specific cost-effective technology. In areas where data availability is scarce, the use of remote sensing data and spatial mapping of available resources can be very useful [9].

In rural electrification planning, there are popular tools like the Network Planner Tool [10] and the Open Source Spatial Electrification Tool (OnSSET) [8,11]. These tools help in planning for meeting electrification targets. OnSSET, for example, has been used by researchers to produce least cost (localized) pathways for meeting 100% electrification targets for selected geographies [12] in Sub-Saharan Africa. However, most energy system models in any of these approaches have largely been limited to techno-economic system details and have rarely incorporated non-technical aspects – i.e. human factors – either social or political. However, energy systems are socio-technical in nature and there have been calls to bring 'realism' into quantitative system modeling, through a deeper engagement with social science [13,14].

There have been few attempts to 'humanize' the modelling frameworks. Li [15] develops an energy system model in the socio-technical framing to simulate human behavior by different system actors. The analysis illustrates the fragility of cost-optimization approaches ignoring behavioral and political factors. Cherp et al. [16] propose a meta-theoretical framework to get an integrated perspective on energy transitions by combining techno-economic, socio-technical, and political aspects. Patankar et al. [17] incorporate conflict uncertainties in energy system optimization model to explore electrification strategies in South Sudan, emphasizing the inclusion of risk in power systems

planning. In the same vein, Korkovelos et al. [18] perform a ‘conflict-adjustment’ to the geospatial energy analysis done for Afghanistan to solve for least cost electricity solutions.

One area that has not been explicitly explored in energy models is application of principles of justice. Technological solutions are prone to exploitation by the existing power structures around them, and they do not inherently produce just outcomes. Rather, justice needs to be explicitly incorporated into planning, implementation and management. In the water sector, for example, there is some evidence within academic literature of use of social justice theorizations for water planning. Joy et al. [19] argue that water problems in India should be framed as justice problems in order to achieve more just distributional outcomes. Lukasiewicz and Baldwin [20] apply the “Social Justice Framework” (SJF) to examine the experiences of different stakeholder groups in water-reform processes in Australia.

Justice has its roots in moral philosophy. Broadly, it relates to the concept of morality (which deals with ‘correct’ behavior) but is narrowly focused on *fairness*. In fact, John Rawls, a well-known 20th century political philosopher who has been hailed for his contributions to the theory of justice, defined justice “as fairness” [21]. He called justice the “first virtue of social institutions” [22]. Going back further in history, justice was termed one of the four cardinal virtues [23].

The principles of justice can be applied to actions, laws and policies. Justice can take many forms. As far as laws and regulations are concerned, it takes a *corrective* and a *distributive* form [24]. Corrective justice is remedial – and it applies when someone’s actions adversely affect someone else. Distributive justice on the other hand relates to the distributions of benefits and burdens across members of the society. While distributive justice cares about fair allocation, the principle of justice broadly relies on some variations of the notion of *equality*. For example, strict egalitarianism, as the name suggests, proposes radical or objective equality with respect to distribution of material goods. Most distributional approaches depart from this and try to incorporate *equity*. Despite having disputed definitions, the most commonly cited definition for equity is the “ratio” concept, based on which, it is defined as the equivalence of outcome/input ratios [25,26]. In the context of distribution or allocation, social psychologist Eckhoff created an equality framework based on different characteristics of the recipients: *need*, *fitness*, *desert*¹, *status*, and *position*. In contrast to objective equality, the framework [27] proposes four more approaches to a fair allocation (Table 1). This study incorporates two of these approaches. One is that of objective equality, and the other is that of relative equality i.e. equity. These two approaches represent the equality and equity scenarios mentioned in the previous section, respectively. Equality approaches consider every recipient equal – and does not consider any of their characteristics. Whereas the equity approach, according to this framework, takes into account recipients’ capabilities (fitness) and whether (or how much) they deserve (desert), based on certain criteria.

The theories of *justice* have also found their way within the purview of energy research, in the form of the increasingly popular but broad conceptualizations of ‘energy justice’. It has its roots in the environmental justice literature. Early environmental justice research focused on concerns around fair socio-spatial distribution of environmental benefits and burdens, i.e. distributive justice. Over last two decades, however, notions around what constitute environmental injustices have diversified, emphasizing the embrace of even recognition and participation [29–31]. Building on these through diverse examples, and emphasizing a geographical (spatial) lens, Walker [32] proposes a three-pronged approach to analyze environmental injustices. First is *distributional*, which, as discussed above, deals with the unequal distribution of impacts and/or responsibilities. Second is justice as *recognition* [33,34], which cover the disrespect and devaluation of various groups in comparison to others. Third lens is of *procedure* [35,36],

Table 1

Principles of equality applied to allocation based on Echhoff’s work [28].

What is to be equal	Relevant characteristics of recipients					
	Need	Fitness	Desert	Status	Position	None
1. Equal amounts to each (objective equality)						X
2. Subjective equality	X		X			
3. Relative equality (equity)		X	X			
4. Rank order equality				X	X	
5. Equal opportunity						X

which deals with the inclusiveness and participation in environmental decision-making. A similar three-pronged approach has inspired ‘energy justice’ as well [37,38]. For example, distributional or distributive justice asks if the costs and benefits of energy changes are fairly distributed. Procedural justice pertains to whether or not relevant stakeholders are included in decision-making. Recognition justice relates to recognizing (i.e. not ignoring) the impacts of energy decisions and those who are impacted by them. The conceptualizations around ‘energy justice’ have been explored in a series of theoretical contributions. Sovacool and Dworkin [39] propose that the concept could act as a conceptual, analytical and a decision-making tool. They define the goal of energy justice as ‘a global energy system that fairly disseminates both the benefits and costs of energy services, and one that has representative and impartial energy decision-making’. They argue for energy decisions to promote the following principles of justice: availability, affordability, due process, good governance, sustainability, inter/intra-generational equity and responsibility. Jenkins et al. [38] promote the application of the three-pronged approach throughout the energy system to identify and resolve various injustices. Other conceptual contributions to energy justice scholarship include perspectives from activism and advocacy [40], how it is uniquely positioned vis-à-vis climate and environmental justice [41], and from the idea of ‘measuring’ justice [42]. All of the aforementioned conceptualizations generally are grounded in the Rawlsian principles of justice². Within the theory of justice, an alternative to Rawls’ ‘justice as fairness’ has been the ‘capabilities approach’ – based on the works of Amartya Sen and Martha Nussbaum [43–45]. This approach considers ‘capabilities’ of the citizens as the main metric of social justice, rather than their incomes or possessions. Day, Walker and Simcock [46] have brought this approach within the purview of energy justice, by conceptualizing and define energy poverty using capabilities framework. They depart from a quantified understanding of energy poverty – as determined by the quantity of energy consumed – and redefine energy poverty as ‘an inability to realise essential capabilities as a direct or indirect result of insufficient access to affordable, reliable and safe energy services, and taking into account available reasonable alternative means of realising these capabilities’.

Most of these theoretical conceptions around energy justice have originated in the context of energy transitions in Western Europe. In fact, they trace back to the notions of fuel poverty and injustice in the UK [47]. Thus, unsurprisingly, the application of these frameworks has also been largely limited to energy issues in developed countries. Liljenfeldt and Petterson [48] explore the distributional justice aspects of windmill siting in Sweden. Fraser and Chapman [49] focus on the similar issue in mega-solar siting in Japan. In the vein of ‘measuring’

² One form of justice not considered in this study is remedial justice in which redistributive efforts are taken to compensate for prior injustices (harm) to specific groups. In the context of electricity access, this could occur if a particular region, for example, was deliberately neglected in prior electrification efforts. However, such remedial justice concerns would be highly locally dependent and likely arise from political pressure and beyond the scope of this study.

¹ A desert-claim means: A “deserves” X by virtue of f.

elements of justice, Chapman, McLellan and Tezuka [50] propose an evaluation framework for energy policy making in Australia. Other studies with their geographic focus on developed countries include: a case study of energy infrastructure of the Netherlands [51], advancing a political ecology of energy justice in Arizona, US [52], and applying the justice framework on the community owned energy generation in Wales [53]. Wood and Roelich [54] use the capabilities framing to analyze the tensions between fossil fuel use, climate change mitigation and human well-being.

Lately, there have been calls within the energy justice literature to learn from a diversity of geographies and energy contexts [55]. Few studies have focused on developing countries as well, by drawing attention to ethical dimensions of energy systems. Alvia-Palavicino and Ureta [56] develop a framework to describe the economization of inequities in energy distribution and apply them to a specific piece of legislation. Islar, Brogaard & Lemberg-Pederson [57] conduct field interviews to assess the feasibility of 'energy justice' in Nepal. Monyei, Adewumi and Jenkins [58] assess the South African non-grid electrification policy from an ethical lens and propose a hybrid energy-generation approach for the energy poor. Malakar [59] uses the capabilities framework to link the efforts of rural electrification in India to people's capabilities and thus argues against the conventional approach of evaluating the benefits of rural electrification – which is by counting the number of electrified households and measuring the quantity of supplied electrified.

By incorporating justice concerns in a modeling exercise, this study demonstrates the utility of quantitative models in solving problems more subjective in nature. Using the tenet of distributive justice, it explores the principle of affordability and intra-generational equity with respect to electricity pricing. However, this contribution does not only have an academic or scholarly value. The electrification tool used in this study, OnSSET, has been actively used by the practitioner community. In fact, WorldBank's Global Electrification Platform (GEP) has also been built using OnSSET [60], with developers and high level decision makers as their main target audience [61]. It is one of the modeling tools promoted for sustainable development, by UNDP and UNDESA [62], and has been used in capacity building for practitioners from many developing countries [63,64]. As an open source tool, it belongs to the growing universe of open source energy modelling tools that are bridging the science-policy gap by evolving through practice and application [65]. Thus, the analytical approach used in this study could directly act as a guide for practitioners involved in electrification planning for poor and developing countries and aiming to incorporate non-technological concerns into their modelling efforts. The geographical focus of this study is Tanzania, which adds to the growing but limited number of studies on energy justice in the context of developing countries. We provide a more detailed description of the geography in the following section.

2. Methods

Here we explain three aspects of the methods used for the study. First, we provide a description and justification for the site chosen for the study. Next, we elaborate on the data analyses methodology, which in essence has two parts – i) the spatial modeling tool utilized (OnSSET) and ii) the analytical approach developed around it.

2.1. Case study description

Sub-saharan Africa suffers from very poor electrification, with close to 600 million still without electricity. It has been argued that renewable energy based decentralized off-grid solutions (both mini-grid and standalone) can help Sub-Saharan Africa achieve its electrification targets [12,66,67]. Tanzania is one of the many countries in Sub-Saharan Africa with very low electrification rate at 32.8% (2017). It has a stable political environment and a growing economy with 6–7% GDP

growth rate for the last decade [68]. It has a public utility called Tanzania Electric Supply Company Limited (TANESCO), which is the sole provider of electricity in the country. Electricity generation and distribution, as well as pricing is regulated by the Energy and Water Utilities Regulatory Authority (EWURA). Tanzania has the total generation capacity of ~1.56 GW. For grid electricity, it relies mainly on hydro (43%) and thermal (57%) power plants [69]. Apart from this, there are 109 mini-grid currently in operation in Tanzania [70], with an installed capacity of ~158 MW (~10% of the total).

2.2. Data analyses

As mentioned in the introduction section, the technological mix for meeting SDG7 is diverse. The centralized grid has to work in conjunction with distributed generation technologies, both mini-grid and standalone systems. Different technology options raise different equity or justice related concerns. Therefore, for this study we want to utilize analytical tools that can capture this diversity of technological options. Moreover, given the spatial and diffuse nature of renewable energy resources as well as of socio-economic equity – geospatial models are well suitable for a study like this. Geospatial models are also advantageous in terms of visualizing the implications of virtue driven policies (like the equity/equality approaches explored in the study), as well as in identifying vulnerable areas. For these reasons, OnSSET is well positioned as a modelling tool because it is not only spatially explicit and technologically rich, it also has been in use by practitioners and planners, as described in the previous section.

The analysis for this paper is built around performing analysis of the outputs produced from OnSSET. A schematic for the overall methodology is shown in Fig. 1 below, followed by detailed descriptions of the three elements: OnSSET, Price modification, and subsidy calculations.

2.2.1. Modeling tool – OnSSET

Taking geospatial and other techno-economic information as inputs, OnSSET produces the least cost technological solution for each location. Fig. 2 shows the data inputs needed and the logical flow of the methodology employed by OnSSET. As the name of the tool suggests, it is open source and is written in the programming language Python. Any freely available Integrated Development Environment can be used to run the algorithms. We used the PyCharm IDE developed by JetBrains (the free version). We want to emphasize that this study is not designed to capture the ground realities of electrification in Tanzania. The study is illustrative, and attempts to address specific questions around electricity pricing and planning using the Tanzania case as a test-bed. For example, one of the limitations of the OnSSET electrification algorithm is that it attributes the entire current electrification to the centralized

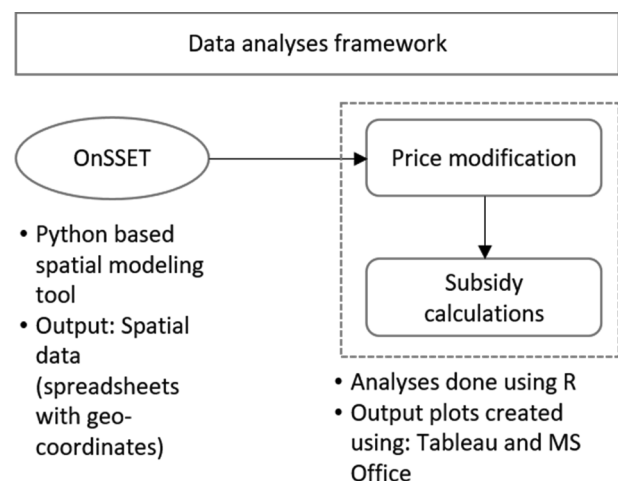


Fig. 1. Data analyses framework used in the study.

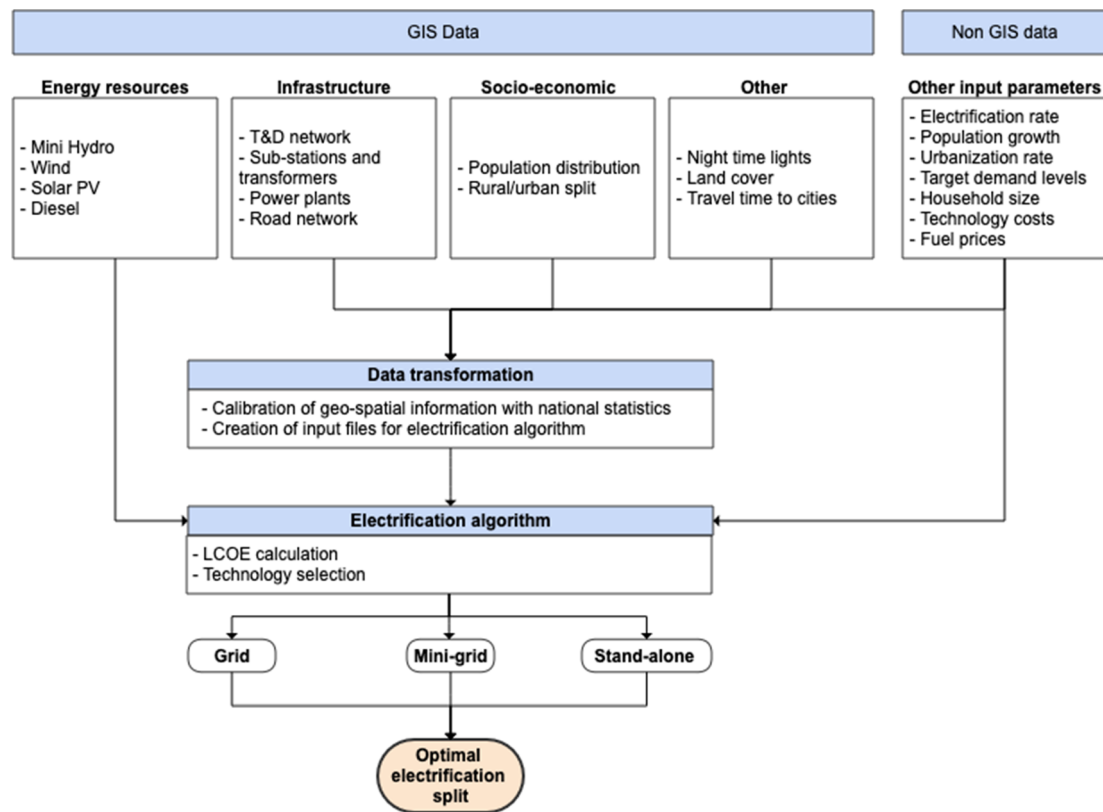


Fig. 2. The OnSSET methodological flow chart. Adopted from Mentis et al. [12] and Korkovelos et al. [71].

grid. This does not capture the reality of current electrification, as the country already has some mini-grid in operation. Having said that, they serve ~184,000 connections together (many of them being commercial connections). While only 17% of these mini-grids are fossil fuel based, they take up 46% of the installed capacity and over 90% of the mini-grid connections. Part of the 109 mini-grids are actually connected to the central grid, and many biomass based mini-grids just serve large commercial customers. However, this study focuses on residential electricity demand by households yet to be electrified. For those, the OnSSET output for Tanzania shows that for 100% electrification by 2030, of the many electrification options technically possible for the country, grid and solar PV based standalone systems are the two main economical solutions. The low penetration of mini-grids and the dominance of just two forms of electrification make interpretation of results for this initial proof-of-principle easier.

Basic demand side modeling parameters and assumptions³ for the algorithm are shown in Table 2. A longer list of parameters is presented in the Appendix A Table A.1. The spatial results from OnSSET include the Levelized Costs of Electricity (LCOEs) of all feasible technologies in each location, and the chosen least cost one. The feasible technologies include: Grid, four mini-grid options (Solar PV, Wind, Hydro, Diesel), and two options for standalone systems (Solar PV, Diesel). The results are in a tabular format where each row represents one grid cell (1 km × 1 km). Under the assumption of cost recovery prices (i.e. zero profit margins), the LCOEs are the same as prices, for all the newly electrified cells. In every grid cell, consumers pay that cost recovery price.

As indicated in the graphical abstract, the main part of the analysis involves modifying or re-distributing electricity prices for the population. We first modify the prices produced by OnSSET, and then proceed to

calculate the subsidies needed to ensure the new price distribution(s).

2.2.2. Price modification

The question of fairness in pricing links to the distribution of these prices. For the prices to be fair, how should they be redistributed? If prices have to be reduced for certain customers, who pays for that subsidy? Where does the money come from and who bears the burden? The methodology used in the study considers a cross-subsidy approach as the starting point for understanding the implications of equality and equity in pricing. We begin with the premise that the entire electric system in the country should be self-financed and actualization of any justice scenario does not impose an extra financial burden on the part of the government. That means a part of the population subsidizes the rest – by having their prices increase. This is a commonly used cross-subsidy model for electricity tariffs [72]. The implications of this purely cross-subsidizing approach in our analysis can then be used as a basis for policy-making, including the possibility of additional financial support from outside the power sector.

The analysis here explores the two approaches of ensuring ‘distributive justice’. The notion of distributive justice assumes a distributing agent and recipients of a service who have claims about what is distributed. In the context of this study, EWURA⁴ – the regulatory agency in Tanzania – is assigned with distributing electricity among the population. The two scenarios used here differ in the criterion used for the distribution. The equality scenario just enforces a constant value for the price across the country. In the equity scenario, we apply the principle of equity by aligning the price with the affordability in that location. As said above, overall revenue generated from electricity consumption is preserved and distributed across the consumers to arrive at differential pricing. Methodologically, this is how the two scenarios are applied:

³ Other detailed list of technology related costs and other parameters has been provided in the Appendix A Table A.1.

⁴ Energy and Water Utilities Regulatory Authority.

Table 2
Assumptions and parameters used for the electrification algorithm.

Parameter	Unit	Value	Source/comment
Start year	–	2018	Model choice
End year	–	2030	Model choice
Population (2015)	million	53.88	UN Population division
Population (start year)	million	59.39	Estimated
Population (end year)	million	83.70	UN population division
Electrification rate (start)	%	32.8	World Bank
Rate of change of urban population	%	2015–2020	UN Population division
		2020–2025	
		2025–2030	
Urban ratio (start year)	%	33.8	UN Population division
Urban ratio (end year)	%	42.2	Estimated
Rural demand	kWh/person/year	160	World Bank Multi-tier framework
Urban demand	kWh/person/year	423	World Bank Multi-tier framework
Discount rate	%	8	Assumed based on trends (Bank of Tanzania)

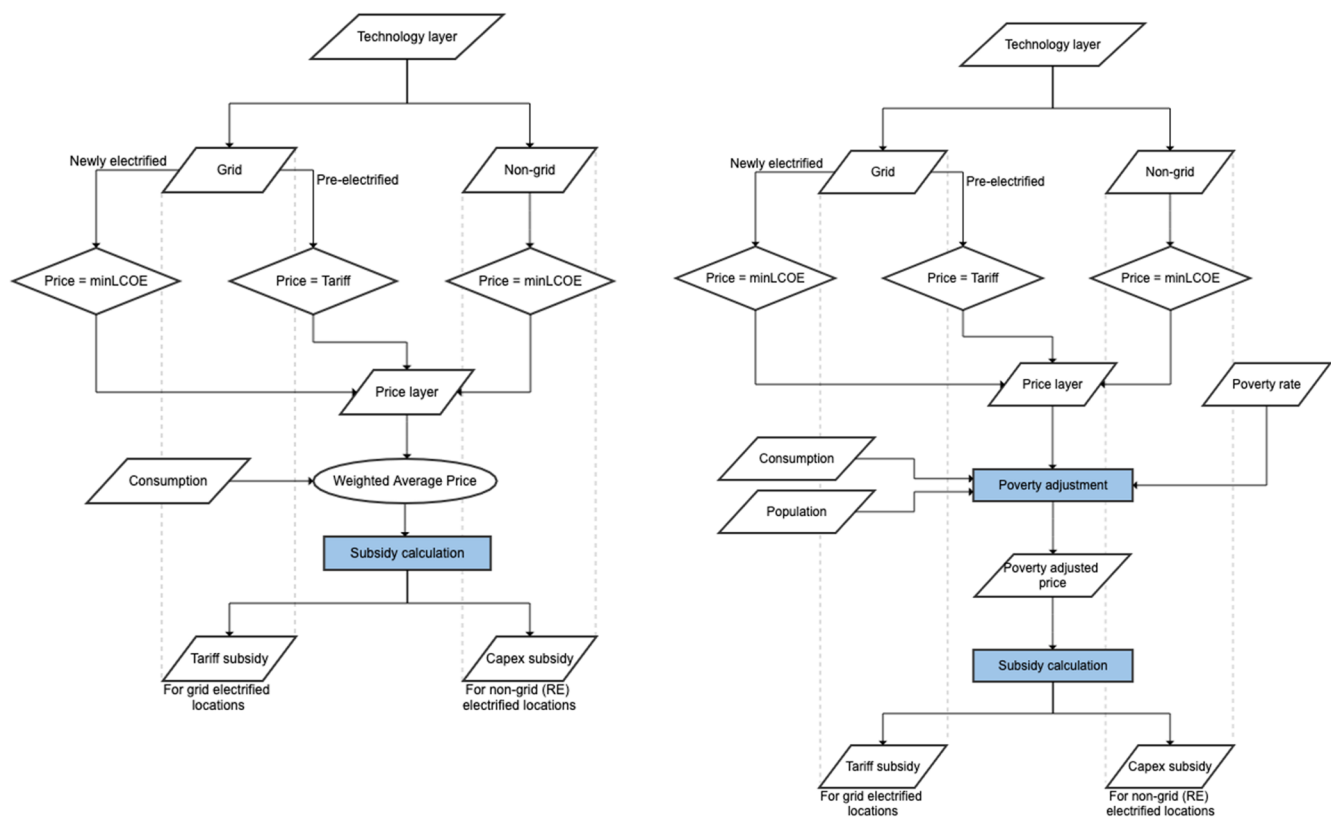


Fig. 3. Analysis methodology for the Equality (left) and Equity (right) scenario. For the equality scenario, there is no poverty adjustment to the price. The price layer is just replaced with a constant value – which is the weighted average price (based on consumption).

- *Equality scenario*: everybody pays the same price for electricity (a weighted average based on consumption and the baseline LCOE)
- *Equity scenario*: price in a certain location is commensurate with affordability of the population in that location
 - o For our analysis – **poverty rate** [73] in that location has been used as a proxy indicator for affordability

We present the methodologies adopted for the two scenarios in the flow charts in Fig. 3. In the equality scenario, after producing the initial price layer, we calculate the ‘desired’ constant price as the weighted average of the prices, using consumption as the weight. This step is different for the equity scenario, where we calculate the desired price layer through ‘poverty adjustment’. A systematic description of the steps involved in the analysis is given below.

1. Price and consumption calculation for the electrified population: price is equal to the existing tariff (US \$/kWh) (as determined by the regulator) and per capita consumption level is calculated based on the current generation and the electrified population.
2. Price and consumption calculation for the remaining population – which is to be newly electrified in the OnSSET analysis: price is equal to the minimum LCOE (as produced by OnSSET). The consumption level for this population is set using the World Bank consumption tiers⁵ – tier 3 for rural locations and 4 for urban.

⁵ International Energy Agency (IEA) and the World Bank. (2015). *Sustainable Energy for All 2015—Progress Toward Sustainable Energy*. <https://doi.org/10.1596/978-1-4648-0690-2>

Box 1**Poverty adjustment.**

$$\text{Revenue} = \sum (\text{Price}_i \times \text{Consumption}_i) \quad (2)$$

$$\text{NonPoorPop}_i = (1 - r_i) \times \text{Pop}_i \quad (3)$$

$$\text{PovAdjPrice}_i = \frac{\text{Revenue} \times \frac{\text{NonPoorPop}_i}{\sum \text{NonPoorPop}_i}}{\text{Consumption}_i} \quad (4)$$

Price_i = price of electricity in grid location i (USD/kWh)

Consumption_i = annual electricity consumption in grid location i (kWh)

r_i = rate of poverty in grid location i

Pop_i = total population in grid location i

NonPoorPop_i = non poor population in grid location i

PovAdjPrice_i = price of electricity after poverty adjustment in grid location i (US\$/kWh)

Price calculation for equality scenario

3. Calculated the weighted average price based on the price and consumption layer.

$$\text{WtAvPrice} = \frac{\sum (\text{Price}_i \times \text{Consumption}_i)}{\sum \text{Consumption}_i} \quad (1)$$

Price_i = price of electricity in grid location i (US \$/kWh)

Consumption_i = electricity consumption in grid location i (kWh)

Price calculation for equity scenario

4. Using the consumption and the price layer, produce the revenue layer (US \$) (Eq. (2) in Box 1).
5. Use the poverty layer to calculate the number of non-poor people living in every grid cell (Eq. (3) in Box 1).
6. Calculate the poverty adjusted price by:
 - a. distributing the total revenue (from the entire population) based on the population of “non-poor” population across locations (Eq. (1) and (2) in Box 1)
 - b. dividing the adjusted revenue by the consumption level of every grid cell (Eq. (3) in Box 1)
7. Create a subset of data by filtering only the locations with price levels higher than the poverty adjusted price. This is done because only these locations would be eligible for any kind of subsidy.

2.2.3. Subsidy calculations

Once the price distributions (either *equal* or *equitable*) have been arrived at, the question is – how should they be operationalized? Financial subsidies have been a common policy instrument to make electricity provision affordable or to promote specific technologies. There have been two primary mechanisms. The first is to subsidize the capital cost of investment (either directly or through subsidies to capital financing). The second is to subsidize the consumption of electricity itself. In principle, subsidy types can be applied irrespective of technologies. In practice, grid electricity (in addition to some preferred financing) has been subsidized through consumption by reducing tariffs for target populations [74,75]. Small scale renewable energy technologies, on the other hand, have generally been supported (if at all) through subsidies on capital expenditure [76]. For simplicity sake, this analysis presumes that tariff subsidies only apply to grid electricity and renewable energy⁶ based mini-grid and standalone systems benefit through capital expenditure (capex) subsidies. The following two subsidy calculation steps are based on this presumption.

1. Tariff subsidies apply to the locations with electrification technology as Grid. It is equal to the difference between the price and the weighted average price or the poverty adjusted price (Eq. (4) in Box 2).

⁶ Diesel technologies (both mini-grid as well standalone) have not been considered for subsidy for this study. As the results show later – the coverage by these technologies is negligible.

2. Capex subsidies apply to the locations with electrification technology as any of the following: mini-grid Hydro, mini-grid Wind, mini-grid PV, standalone PV. As the price for these locations was equal to the minimum LCOE, we ensure a poverty-adjusted price by making an adjustment in the LCOE (new LCOE), which in turn can be ensured by a certain amount of capex subsidy. We calculate the amount of this subsidy by using the LCOE equation (Eq. (6) in Box 2).

Commensurate with the methodology presented above, the analysis produces two levels of relevant outputs, for the two scenarios. First is the price(s) to ensure the corresponding principle of justice. Second, is the subsidy pattern required to ensure that price(s). We discuss these outputs in the next section in detail, along with their implications. Additionally, as the description above shows, the outputs are highly contingent on the initial assumptions and input conditions. That is, changing input values/targets might indicate a different mix and LCOE distribution. However, the precise technology mix is not really the main scope and purpose of this study, but we have performed a sensitivity analysis (Appendix B) to incorporate a variety of input conditions. Having said that, this analysis compares results for different *scenarios* rather than testing the impact of *uncertainties*. In fact, this has motivated the use of OnSSET for scenario analysis, e.g. in Malawi (6 scenarios) [71] or for the Global Electrification Platform (200+ scenarios) [60]. In this context, we have just selected a few relevant parameters (urban ratio, discount rate, demand tiers) to define five different sensitivity scenarios (Appendix B).

3. Results and discussion

We first present the output of the baseline case – the technology mix for Tanzania to achieve 100% electrification by 2030 and the corresponding LCOE distribution. This is followed by the results about how prices are impacted – i.e. the redistribution of prices based on the two scenarios and the resulting subsidy distributions. Subsequently, we discuss the implications of the findings for a variety of stakeholders.

3.1. Baseline electrification results

Before we discuss the redistribution results, it is important to look at the baseline electrification picture for Tanzania, as produced by the OnSSET electrification algorithm. Fig. 4 shows the baseline technological mix, segregated between rural and urban population. In 2030, the majority of the population (78%) gets electricity from the centralized grid. Among off-grid technologies, only standalone PV systems have any substantial contribution, covering 22% of the population – all of it in rural areas. It is clear, that in terms of cost of electrification technology, the grid remains the least cost alternative for all the urban areas, and a large segment of rural ones. Compared with the similar work done for other countries in the region like Nigeria [11] and Ethiopia [8], Tanzania has much less diversity in the electrification mix. There is marginal contribution by the other technologies as well, especially PV and wind-based mini-grid and standalone diesel

Box 2**Subsidy calculation.**

If $Price_i > PovAdjPrice_i$

For locations getting Grid electricity.

$TariffSubsidy_i = Price_i - PovAdjPrice_i$ (4)

For locations getting renewable energy based non-grid electricity, capital expenditure subsidy is calculated based on the following:

$(new)minLCOE_i^* = PovAdjPrice_i$ (5)

$$minLCOE_i^* = (CapexSubsidy_i * I_i) + \frac{\sum_{t=1}^n \frac{O \& M_t}{(1+d)^t}}{\sum_{t=1}^n \frac{En_t}{(1+d)^t}} \quad (6)$$

$TariffSubsidy_i$ = requiredtariffsubsidy in grid location i (USD/kWh)

$minLCOE_i^*$ = newminimumLCOE in grid location i (USD/kWh)

$CapexSubsidy_i$ = required capex subsidy in grid location i (%)

I_i = technology investment in grid location i (USD)

$O \& M_t$ = operation and maintenance cost for the technology in year t (USD)

d = discount rate

En_t = energy produced in year t (kWh)

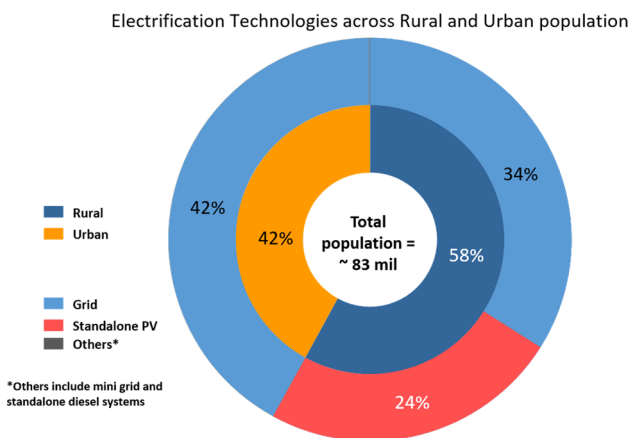


Fig. 4. Technology mix for electricity as per OnSSET for Tanzania in 2030 – in the form of a sunburst plot. The grid electrifies the majority of the population. In rural Tanzania however, standalone PV systems make a significant contribution.

systems. However, we want to emphasize that the results are also contingent on the selected set of input variables, and different input variables produce different OnSSET scenarios (as indicated above). According to the sensitivity analysis performed, four out of five scenarios do not show any perceptible change in the distribution of technologies. Lowering the rural household demand level however (sensitivity scenario 5) seems to show a significant change in terms of population distribution of technologies as well as subsidies. This is due to a higher section of the population moving from grid electrification to standalone PV. The spatial map of electrification technologies for Tanzania has been shown in Fig. A3. As mentioned in the methods section, for this analysis LCOEs correspond to the price of electricity. The baseline distribution of prices ranges from about 11 cents to less than 50 cents (The distribution is shown later in Fig. 5).

3.2. Justice and electricity prices

We show the impact of the justice approaches on baseline price distribution using three sets of plots – Figs. 5–7. Fig. 5(a) shows the price distribution in terms of population in the baseline and the two scenarios – although the equality scenario price is just constant at a weighted average value. As clear from a comparative look between, the poverty adjusted price distribution (equity scenario) is more dispersed than the original (baseline) price distribution. After poverty adjustment, the median price comes down from US\$0.38 to around US\$0.22. Poverty adjusted prices range all the way from a few cents to as high as US\$0.70. Fig. 5(b) shows the price distribution in terms of number of locations at a particular price level.

In the baseline, sparsely populated areas pay higher price for electricity. Most of these areas get electricity from standalone systems (red bars). In the equity scenario, many of these red bars move towards the left – i.e. they see their prices reduce. At the same time, many densely populated areas served with grid electricity (blue bars) see their prices increase – the bars move towards the right (post poverty adjustment) as compared to their baseline positions (left plots). This is further visible in Fig. 6 as well.

Fig. 6 shows how the two approaches compare in terms of share of population experiencing price change. In the equity scenario, a smaller segment of the population experiences price increase as compared to the equality scenario (46% vs. 48%). This means that 54% of the population (~45 million people) receive a subsidy (i.e. price reduction) in the equity scenario, as compared to 52% in the equality scenario (~43 million). The demographic impact of the two scenarios is also different. The benefits of price reduction (subsidy) in the equality scenario are limited to rural Tanzania. Whereas in the equity scenario, of the 45 million (54%) people receiving subsidies, around 8.3 million – ~10% of the total population – belong to urban areas. The flip side is that these benefits are drawn from a higher burden on the rural population. Among those who see their prices increase in the equity scenario, 14% belong to rural areas as opposed to 6% in the equality scenario. Due to the poverty adjustment, areas with lower rates of poverty (whether rural or urban) are subsidizing those with higher poverty incidence.

The plots in Fig. 7 show the distributions and ranges of price changes in the two scenarios. Positive value represents increase in prices and a negative value represents price reduction. In the equality scenario (7(a) and (b)), the prices reduce by up to US\$0.30/kWh. For the equity scenario (7(c) and (d)), the maximum value of price reduction is higher, but so is the price increase. For some locations, prices increase by over US\$0.40/kWh. On average, we see price levels coming down. In the equality scenario, only areas served with grid electricity, witness price increase. Most of these locations are densely populated (i.e. urban), as indicated by very high blue bar in 7(b). Whereas in the equity scenario, we see a few red bars with positive values as well, meaning that even some areas served with standalone systems, see their prices increase. These locations are sparsely populated (low population bars in 7(d)). However, some densely populated grid electrified areas also benefit from price reduction – as visible in the blue bar distribution around zero in 7(d). As expected (based on 5(b)), price change also follows a normal distribution in the equity scenario.

3.3. Subsidy distributions

In our analysis, RE based DG technologies (mini-grid and standalone) are eligible for capital subsidies, whereas the grid connections are eligible for tariff subsidies.

We first show the capex subsidy plots for the two scenarios in Fig. 8. The points in each scatter plot represents one location. A big difference

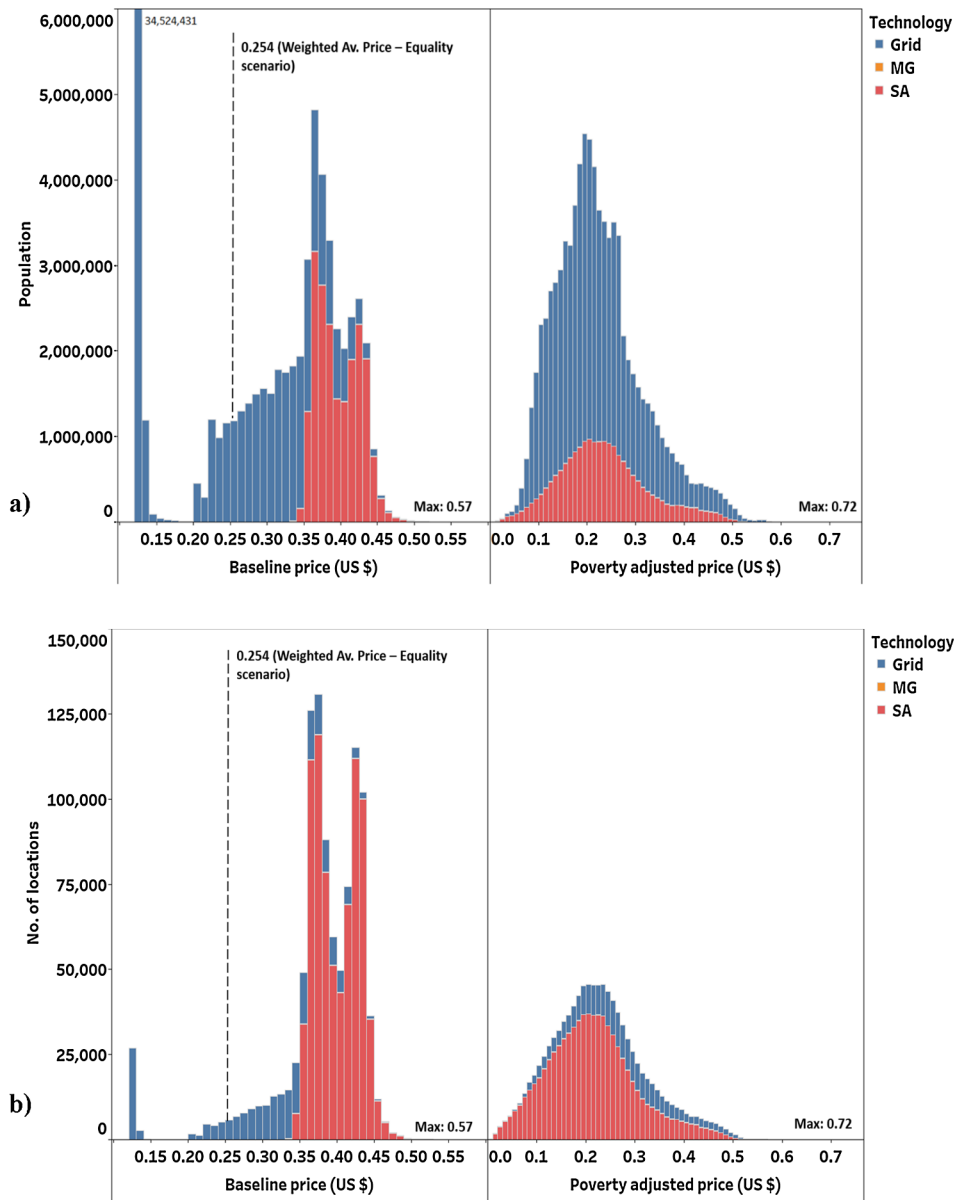


Fig. 5. Distribution comparisons between baseline price, equality scenario price (weighted average) and poverty adjusted prices (equity scenario). The bar height represents the population in (a) and the no. of locations in (b). Poverty adjusted prices follow a near normal distribution, irrespective of the electrification technology. It covers a wider range as well, as compared to the baseline prices. One thing to note here is oversized bar in the baseline price. 34 million ((a) left) pay a fixed tariff rate of US\$ 12 cents in the baseline, as per the model design. The color codes correspond with the type of electrification technology. MG = Mini-grid; SA = Standalone system.

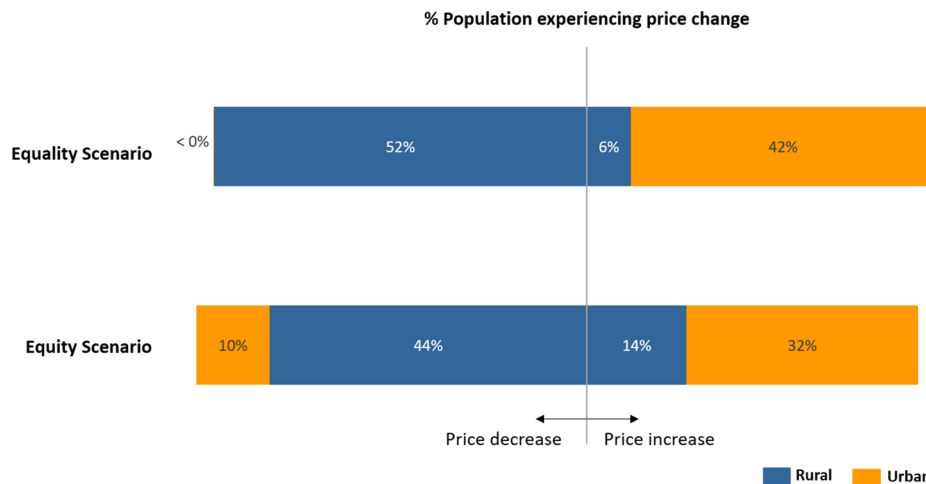


Fig. 6. Comparison between the two scenarios in terms of their effects on prices. The most significant difference between the two scenarios is that in the equity scenario, a significantly higher segment of rural population witnessed price increase, as compared to the equality scenario.

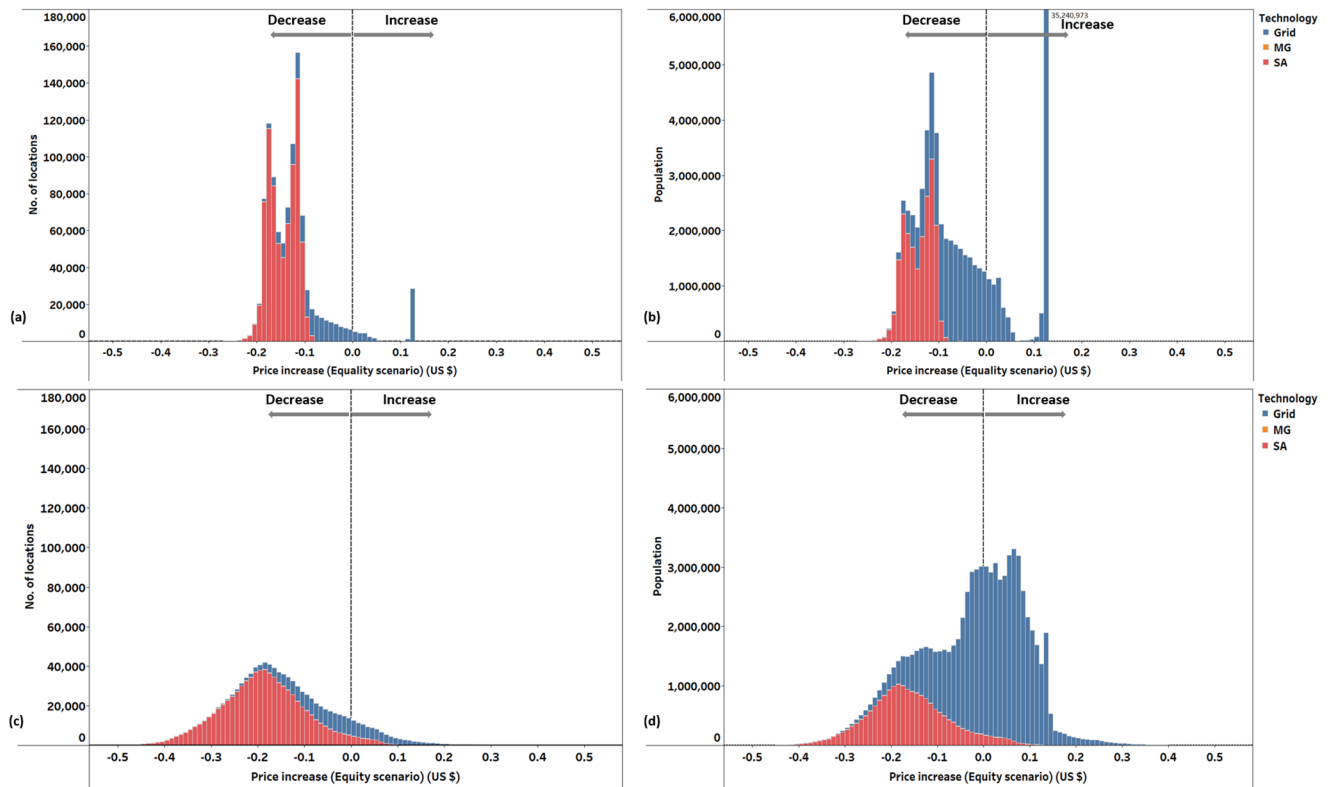


Fig. 7. (a) Frequency distribution for price change in the equality scenario ($\text{Price increase} = \text{WtAvPrice} - \text{Baseline Price}$). The bar represents the number of locations for a specific value of the price increase (or decrease). (b) Price increase is plotted against the population for the equality scenario. One point represents one location (grid cell). ((c) and (d)) Corresponding plots for the equity scenario ($\text{Price increase} = \text{Poverty adjusted price} - \text{Baseline Price}$). The color codes correspond with the type of electrification technology. MG = Mini-grid; SA = Standalone system.

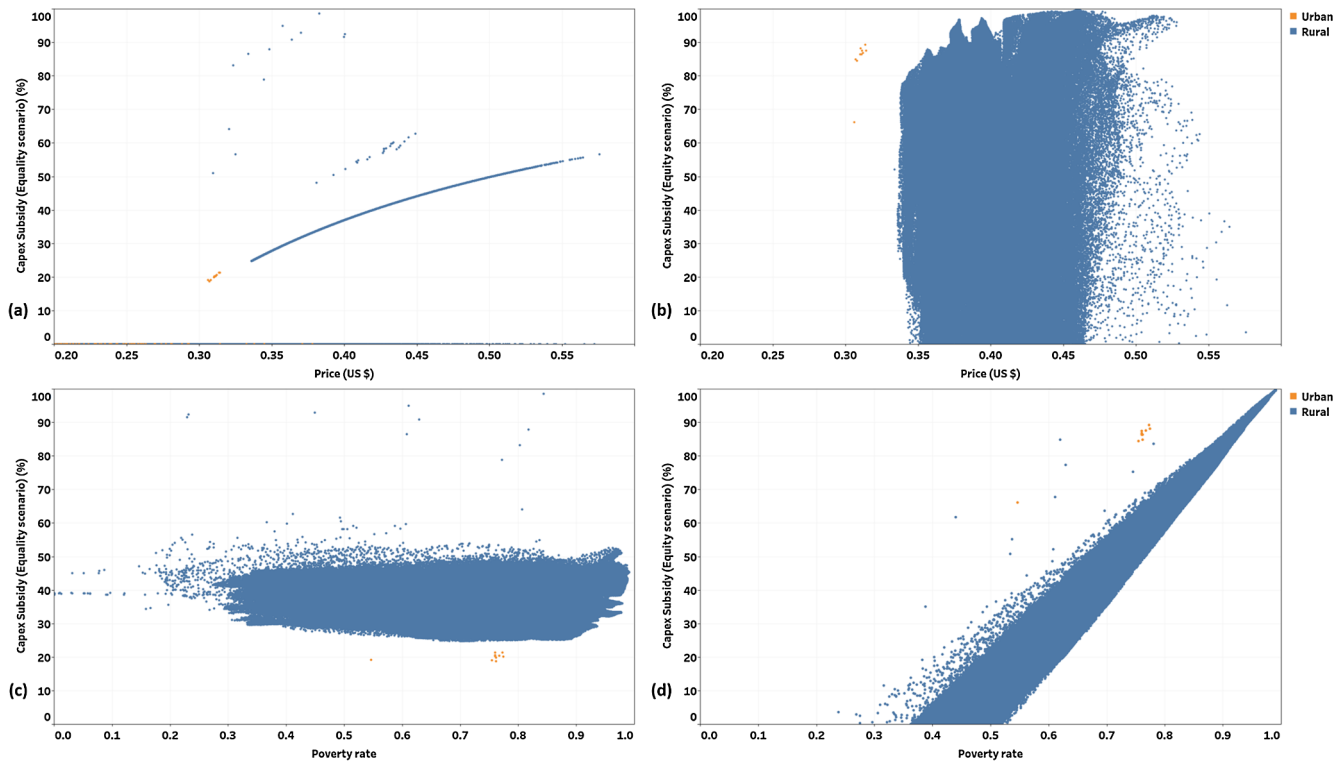


Fig. 8. Capital expenditure (capex) subsidy outputs plots against baseline price and poverty rate. (a) and (c) are for equality scenario. (b) and (d) are for equity scenario. Each dot represents one location. The baseline price range shown here is between US\$ 0.20 to US\$ 0.60. Color-coding is meant to differentiate between the urban and rural locations.

between the two scenarios in terms of their capex subsidy outputs is the range. In the equality scenario (8(a) and (c)), subsidy values for almost all the eligible locations (~99.9%) fall within the range of 25–50%. Whereas, as both 8(b) and (d) indicate, the equity scenario covers the entire range of 0–100%. There are a few locations in both scenarios that have capex subsidy values well beyond 100% (not shown in the plots). These locations have mini-grid as their electrification technology rather than standalone systems. Values over 100% capex subsidy mean that even 100% rebate on the capital expenditure is not enough for these locations to make electricity prices affordable.

Fig. 8(a) shows a clear positive relationship between the baseline price and the capex subsidy in the equality scenario. Locations paying more in the baseline get a higher subsidy. In this scenario, we see an absence of any relationship between subsidy values and poverty rate 8(c). In the equity scenario however, there exists a pattern (8(d)) between the amount of subsidy available and poverty rates (as would be expected). Areas with higher percentage of poor population get a higher subsidy. However, at the same time, this scenario does not honor the baseline price level in deciding the subsidy level (as evident in 8(b)). It is interesting to note across the four plots that even though there are only few urban locations (orange dots) getting capex subsidy in either scenario, in the equity scenario they are able to avail a much higher value (80–90%) as compared to the equality scenario (20–30%).

Similar patterns exist for tariff subsidies (Fig. 9), with two interesting differences. First, a significant number of urban locations benefit from a tariff subsidy in the equity scenario than in the equality scenario. Second, a positive relationship between (tariff) subsidy and the baseline price is visible also in the equity scenario (9(b)) – which is not the case for capex subsidy (8(b)). This is due to how the two subsidies are calculated (Ref. Box 2 in the methods section)

The cumulative density plots in Fig. 10 show the scope of subsidy benefits for the two scenarios. The geographical coverage (no. of locations) and the population coverage of capex subsidy is marginally higher for the equality scenario (Fig. 10(a)). Subsidy values for the equity scenario follow a S-curve. Aligning well with Fig. 8(a), the

middle plot in Fig. 10(a) shows that most of the population falls within the range of 30–50% of capex subsidy in the equality scenario. In addition, just as in Fig. 8, not many urban locations benefit from capex subsidies in either scenario.

Fig. 10(b) plots show interesting contrasts between the two scenarios in terms of their tariff subsidy outputs. The range of the tariff subsidies in the two scenarios also differ. The highest value of tariff subsidy is US\$0.42 in the equity scenario while the maximum subsidy in the equality scenario is only US\$0.32. An important difference between the two scenarios is a significantly larger presence of urban beneficiaries in the equity scenario. As these locations are densely populated, the population coverage is higher even though the geographical reach (no. of locations) is lower in the equity scenario. Twenty eight percent of the total population benefits from tariff subsidies in the equality scenario as opposed to 32% in the equity scenario (10(b) middle). With respect to benefits to the poor population, the equity scenario reaches ~8% more poor people than the equality scenario (10(b) top). As far as the sensitivity analysis (Appendix B) is concerned, the core findings about equity vs. equality do not materially change across four of the five sensitivity scenarios we create. The fifth scenario – expectedly shows slightly different results because of a very different technology mix. However, as previously described – the sensitivity testing is akin to a scenario analysis.

3.4. Implications of above findings (for different stakeholders)

3.4.1. For developers

Off-grid and mini-grid developers worry about technology cost, not just their own but also the competing technologies. Customers' ability to pay matters for investment decisions as well. They cannot pass on very high costs to the end consumer. This OnSSET based analysis covers the following aspects of their concerns:

- The spatial variation of technology costs throughout the country.
- The spatial variation of customer's ability to pay (affordability).
- Possible financial support structures available to regulatory bodies

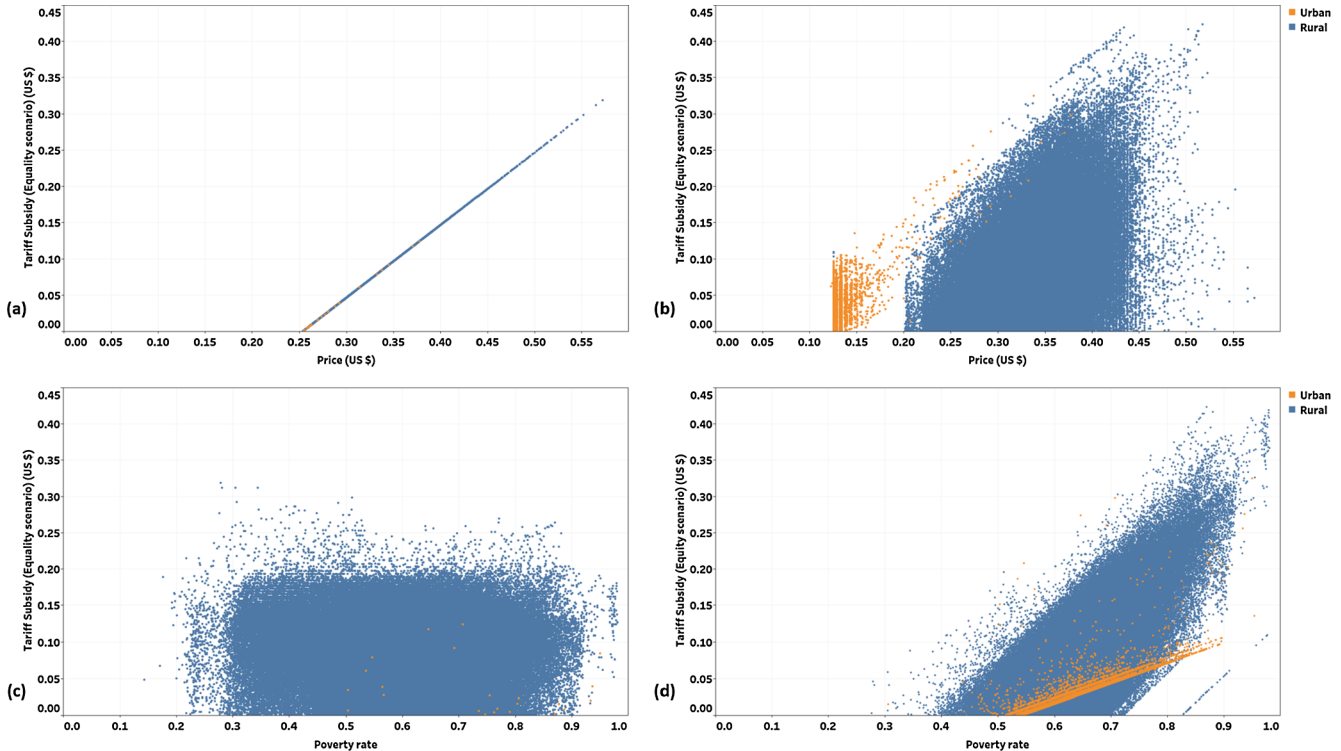


Fig. 9. Tariff subsidy outputs plots against baseline price and poverty rate. (a) and (c) are for equality scenario. (b) and (d) are for equity scenario. Each dot represents one location. The baseline price range shown here is between 0 to US\$ 0.60. Color-coding is meant to differentiate between the urban and rural locations.

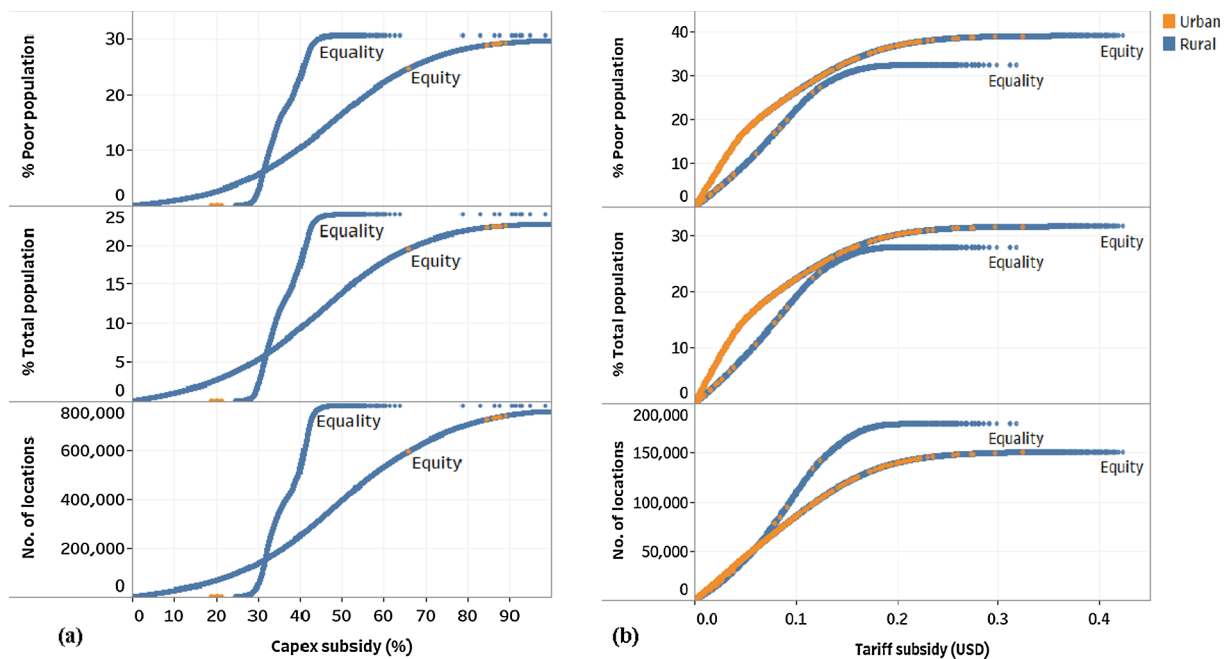


Fig. 10. Cumulative density plots for (a) capex subsidy and (b) tariff subsidy, respectively. Subsidy values are plotted against the total number of locations (bottom), % of total population (middle), and % of poor population (top). Each dot represents one location. Equality and equity scenario trends are shown distinctly, and the color-coding differentiates between urban and rural locations.

in order to both help investors recover costs and address affordability concerns.

Grid electricity relies on high customer density and high consumption to be competitive. In populated urban areas therefore, grid emerges as the least cost electrification option. In remote and sparsely populated areas, solar PV based standalone systems are the cheapest among all options. In certain peri-urban areas, mini-grids are able to outcompete all others due to moderate cost and sufficient customer volumes.

The poverty distribution, together with the cost distribution (from OnSSET), speak to the developers' sense of risk. They may deem areas where poverty is widespread and costs are higher, to be riskier to invest in, due to poorer affordability. This analysis tells us how developers (for both the grid and non-grid technologies) could rely on complex capital and tariff subsidy plans to recover their costs without overcharging the end-users.

As mentioned in the results, some of the capex subsidy values cross 100%. These locations are all electrified by mini-grid technologies as the least cost technology. What is special about the locations that even a 100% capex subsidy is insufficient to ensure cost recovery? These areas are peri-urban but far enough from the centralized grid to make grid extension expensive, but dense enough by population to make mini-grid feasible (relative to standalone systems). In such locations, even the cheapest technology option is very expensive. This is particularly a problem in areas where poverty rates are also high.

3.4.2. For end users

The equality scenario remains agnostic towards consumers' ability to pay, and only cares for technology costs. This means that urban locations, where technology costs are generally lower end up seeing their prices increase irrespective of their affordability levels. But the equity approach also considers people's ability to pay. That is why we see even urban locations benefitting from price drops (orange dots in Figs. 9 and 10). The flip side to this is that some rural areas also bear the burden of price increases in the equity scenario, because they have lower levels of poverty. In essence, vulnerable areas (locations with higher incidence of poverty) stand a higher chance of receiving benefits in terms of subsidy support.

Based on the analysis, it can be argued that the equity scenario is better than the equality scenario in terms of beneficiary population (how many

people benefit) as well as subsidy burden. More people are rewarded (price reductions) in this scenario as compared to the equality scenario. Also, less people are affected by price increase in the equity scenario (38 million versus 40 million). Per capita burden of price increases, is significantly less in the equity scenario – at US\$23 per year – as compared to the equality scenario, which is at US\$48 per year. In both the scenarios, it is mostly the grid customers that bear the burden of price increase. However, average price increase in grid electrified locations is ~36% in the equity scenario, as compared to 30% in the equality scenario. At the same time, grid connected consumers enjoy a higher average annual savings (due to tariff subsidy) in the equality scenario (US\$12 versus US\$8 in the equity scenario).

3.4.3. Implications for policy makers

In terms of ease of (policy) implementation, adhering to the simple *equality* principle – in which every location is treated equally, is much simpler to implement as compared to the *equity* scenario. The former proposes one common electricity price for everyone. The latter demands a unique price for every location. Also, as the capex subsidy results show, the range of subsidy values in the equality scenario is much narrower as compared to the equity scenario. It may arguably be considered easier to choose from a narrower set of values than a range which is spread out between 0 and 100%.

Justice is a politically sensitive issue as well. The justice goals we analyze rely on implementing cross-subsidy models. That is, one part of the population is subsidizing the rest. Some people are over-charged, so that others can have electricity for cheaper. The amount of price increase, and the demographic background of those who bear it – can be crucial to the feasibility of any policy. In the equality scenario, only densely populated areas (urban) are bearing the burden of price increase. The equity approach results in a fairer (or more just) distribution – it spreads that burden, but in doing so – even remote locations that are not necessarily poor end up seeing their prices go up to a much higher level than in the equality scenario. Thus, the political risks of the two approaches are different, and the equity approach may be considered riskier, given that the prices increase for a part of the rural population. However, there are ways to mitigate these risks. Rather than seeking to generate subsidy amounts by increasing prices for a segment of the population, the same can be done via outside support – e.g. from the general budget, or through external debt.

4. Conclusion

In this study, we demonstrate a new application of a spatial electrification tool in exploring questions of *justice* in energy planning in developing countries. Although the analysis is more of an illustrative exercise, it has potential to inform policy and investment decisions. A number of stakeholders can draw important insights by viewing the outputs of this study from their standpoint. More specifically, we explore implications of these findings for developers, consumers and policymakers (as discussed in the above section).

The analysis leaves a lot of room for maneuvering in the methodology itself. For example, the adjusted price (equity scenario) and subsidy distributions follow the distribution of poverty rate (refer to Figs. A1 and A2). This is due to the fact that poverty rate is used as a proxy for affordability in this analysis. How well the poverty rate approximates the affordability of a location is arguable. Ideally, spatial income data should be used to arrive at some direct indicator of affordability. Although income data is hard to obtain, in future work, we hope to use a better approximation of affordability. Apart from more or better data, we wish to explore the possibilities of ground-truthing the analyses by incorporating stakeholder views or public perceptions into the model. However, even with the use of poverty rate, there is scope for adding more layers to the analysis. For example, using the same poverty rate – we can do split pricing for any particular grid location. One price for low income people, and another for the rest. Alternatively, as we see prices shoot up as high as over US\$ 0.70/kWh cents for some locations, one approach could be to put an upper cap to the allowable price increase. This would mean a reduced revenue generation for cross-subsidization. What could this mean for the resulting financial burden on the part of the government? Apart from methodological alternatives, in our analysis, the way we treat different technological solutions also was unique. We ignored the areas that get electrified by diesel generation. Do they not deserve any financial

rebate? The methodology could include them by considering fuel subsidies. This study thus is the first in a series of analyses exploring these different questions. In future analyses, we plan to revise the methodological scope by considering standalone and mini-grid diesel systems in the subsidy analysis and by incorporating more sophisticated approaches to *justice*. It would also be interesting to see how this justice related analysis plays out in other contexts.

CRediT authorship contribution statement

Vikas Menghwani: Conceptualization, Methodology, Formal analysis, Software, Data curation, Writing - original draft, Visualization. **Hisham Zerriffi:** Supervision, Resources, Funding acquisition, Conceptualization, Methodology, Writing - review & editing. **Alexandros Korkovelos:** Resources, Software, Writing - review & editing. **Babak Khavari:** Software, Writing - review & editing. **Andreas Sahlberg:** Software. **Mark Howells:** Conceptualization, Resources, Funding acquisition, Writing - review & editing. **Dimitris Mentis:** Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

See Figs. A1–A3 and Table A1.

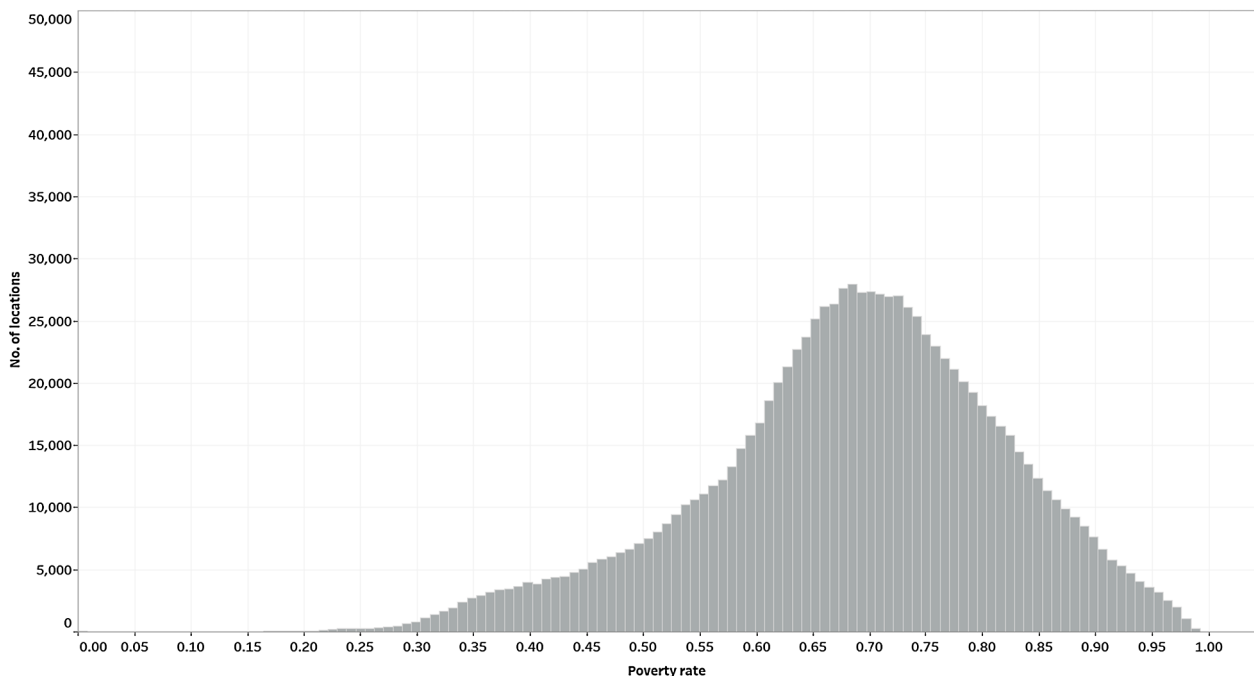


Fig. A1. Poverty rate histogram (Source: Worldpop [74]). 1 location refers to one 1 km² grid cell.

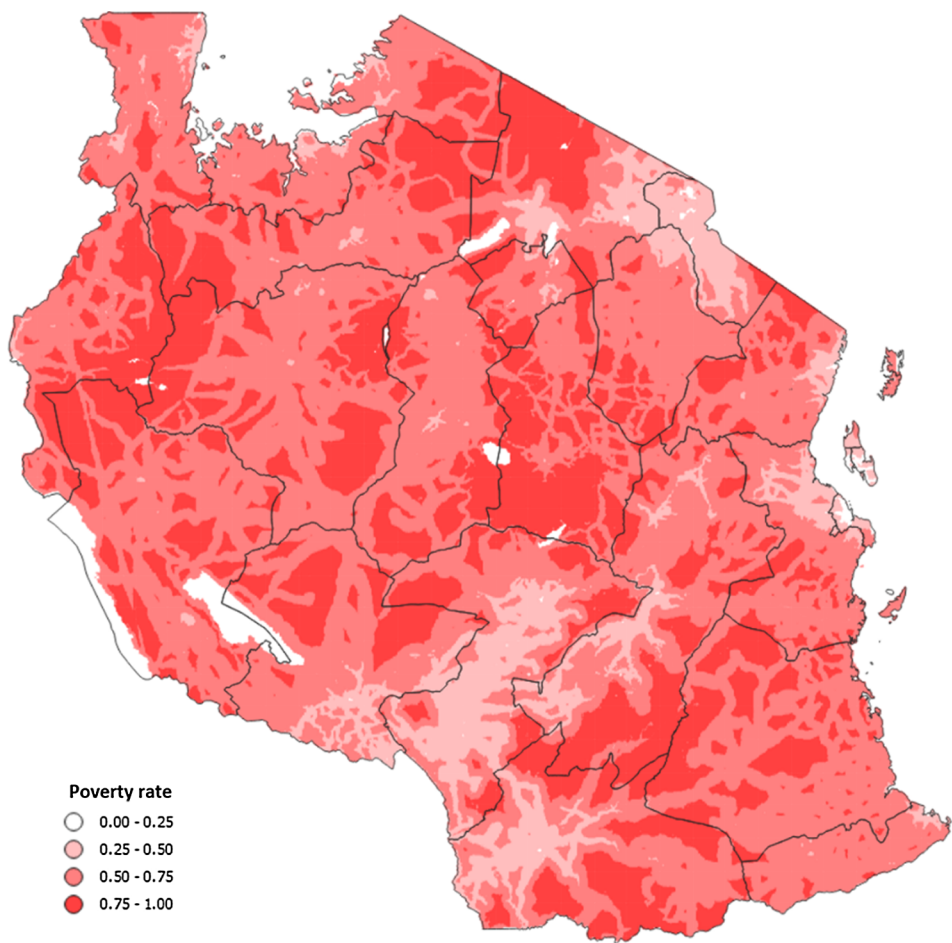


Fig. A2. Poverty rate across Tanzania (Source: Worldpop [74]).

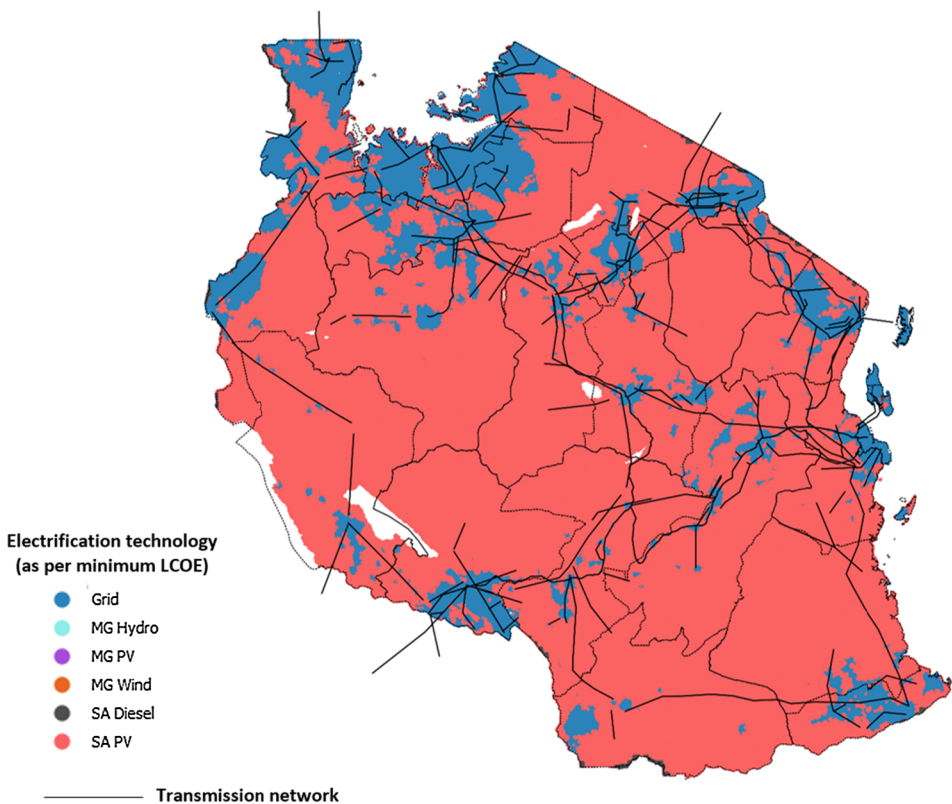


Fig. A3. OnSSET output in terms of technologies needed to electrify the entire country. The technology is chosen based on the LCOEs – the one with the minimum value for each location is selected.

Table A1
Values for technical parameters used in OnSSET.

Parameter	Value	Unit
Start year	2018	
End year	2030	
Hours per year	8760	Hours
Grid cell area	1	km ²
MV line cost	9000	US \$/km
LV line cost	5000	US \$/km
MV line capacity	50	km
LV line capacity	10	km
LV line max length	30	km
HV line cost	53,000	US \$/km
MV line max length	50	km
MV increase rate	10%	%
Grid connection cost (per household)	125	US \$
Grid cost ratio	10%	%
Mini grid O&M for T&D lines	3%	%
Mini grid distribution losses	5%	%
Mini grid Solar PV Base to Peak ratio	0.9	Ratio
Mini grid Hydro Base to Peak ratio	0.5	Ratio
Mini grid Wind Base to Peak ratio	0.75	Ratio
Mini grid Diesel Base to Peak ratio	0.5	Ratio
Mini grid Solar PV capital cost	4300	US \$/kW
Mini grid Hydro capital cost	5000	US \$/kW
Mini grid Wind capital cost	3000	US \$/kW
Mini grid Diesel capital cost	721	US \$/kW
Mini grid connection cost (per household)	100	US \$
Mini grid Solar PV technical life	20	Years
Mini grid Hydro technical life	30	Years
Mini grid Wind technical life	20	Years
Mini grid Diesel technical life	15	Years
Mini grid Hydro capacity factor	0.5	Ratio
Mini grid Diesel capacity factor	0.7	Ratio
Mini grid Solar PV O&M cost	1.5%	%
Mini grid Hydro O&M cost	2%	%
Mini grid Wind O&M cost	2%	%
Mini grid Diesel O&M cost	10%	%
Discount rate	8%	%
Standalone Solar PV technical life	15	Years
Standalone Solar PV capital cost	5500	US \$/kW
Standalone Solar PV Base to Peak ratio	0.9	Ratio
Standalone Solar PV O&M cost	1.2%	%
Standalone Diesel technical life	10	Years
Standalone Diesel capital cost	938	US \$/kW
Standalone Diesel Base to Peak ratio	0.5	Ratio
Standalone Diesel O&M cost	10%	%
Standalone Diesel Efficiency	28%	%
Mini grid Diesel Efficiency	33%	%
Diesel price	0.92	US \$/litre
Truck volume (Standalone systems)	300	Litres
Truck consumption (Standalone systems)	14	km/litre
Truck volume (Mini grid)	15,000	Litres
Truck consumption (Mini grid)	33.7	km/litre
Lower Heating Value (LHV)	9.944549	kWh/litre

Appendix B. Sensitivity analysis

We have selected a few relevant parameters to create five different sensitivity scenarios. Through this analysis, we check to see if change in values of specific input parameters, leads to a significant change in the output results, on which findings are based. We have checked sensitivity on three specific input conditions: discount rate value, future urban ratio value, and the electricity demand tiers for rural consumers. The following table shows the scenarios are described below:

1. Sensitivity scenario 1 (SS1): **Future urban ratio increases by 10%**
2. Sensitivity scenario 2 (SS2): **Future urban ratio decreases by 10%**
3. Sensitivity scenario 3 (SS3): **Discount rate changed to 10%**
4. Sensitivity scenario 4 (SS4): **Discount rate changed to 6%**
5. Sensitivity scenario 5 (SS5): **Rural demand tier changed from 3 to 2**

The scenarios above test the sensitivity of the two sets of findings (Ref. Fig. 1). First is of the OnSSET output – which includes the technology mix and the consumer population for each technology. This is shown in Table B.1. The second pertains to the core findings of the study – which are comparative analyses of the two justice approaches (equality and equity). We check a few specific indicators for the two justice approaches for each of the above sensitivity scenario. Rather than emphasizing the actual figures for the two approaches – we try to compare the scenarios with the study

model in terms of the relative performance of the two justice approaches on specific indicators. A simplified table is then produced in Table B.2. For reference, the numbers are included in Table B.3.

Table B.1 shows that a change in the urban ratio (SS1 and SS2) by 10% leads to change of ~6% in the overall capacity requirement for electrification. An increase in urban ratio, by way of increase in overall demand, leads to more capacity. Most of the impact is on grid electrification. Expectedly, SS5 shows similar effect on capacity. Choosing a lower demand tier for rural population reduces the overall capacity needed for electrification, as well as move many more people over from grid to standalone PV. Changes in discount ratio (SS3 and SS4) – have negligible effect on OnSSET output. Marginal effect is seen when an increase in discount rate shifts some customers from standalone PV to diesel.

As per Table B.2, the relative performance of the equality and equity approaches does not really change across the sensitivity scenarios. For example, a much larger urban population gets subsidy benefits in the equity approach in the study model, and this finding remains the same throughout the sensitivity scenarios. Equality approach leads to lower annual per capita savings than the equity approach – this also does not change throughout the scenarios. Having said that, SS5 does flip the findings (red) for some of the indicators. For instance, for SS5, more people bear the burden of price increase in the equity approach than equality one. Consequently, less people benefit from subsidy in the equity approach in this scenario. This is not surprising. As more people in rural areas get electricity from standalone solar PV, equality approach benefits (in terms of subsidy) a large rural population as compared to the equity approach.

Table B1

The table compares the five sensitivity scenarios with the original (study model) in terms of the OnSSET outputs they produce. Only SS5 shows a significant change as compared to the distribution of the customer population across technologies.

Technology mix (OnSSET)	Study model	SS1	SS2	SS3	SS4	SS5
Total new customers (million)	64.41	64.41	64.41	64.41	64.41	64.41
New customers (Grid %)	69%	69%	68%	69%	69%	25%
New customers (Standalone PV %)	31%	30%	32%	31%	31%	74%
New customers (Others %)	< 1%	< 1%	< 1%	< 1%	< 1%	< 1%
Total new capacity (MW)	5843	6169	5517	5832	5842	3641
New capacity (Grid %)	71%	73%	68%	71%	71%	69%
New capacity (Standalone PV %)	29%	27%	32%	29%	29%	31%
New capacity (Others %)	< 1%	< 1%	< 1%	< 1%	< 1%	< 1%

Table B2

The table compares the sensitivity scenarios with the original scenario (study model) with respect to the core findings about the justice approaches analyzed in this study. SS1 to SS4 do not lead to any changes in the relative performance of the two approaches. SS5 produces different results by some indicators.

Subsidy related indicators	Equality > / < / = Equity					
	Study model	SS1	SS2	SS3	SS4	SS5
Price increase (total population)	>	>	>	>	>	<
Price decrease (total population) i.e. subsidy beneficiaries	<	<	<	<	<	>
Price increase (urban pop %)	>	>	>	>	>	>
Price increase (rural pop %)	<	<	<	<	<	<
Price decrease (urban pop %)	<	<	<	<	<	<
Price decrease (rural pop %)	>	>	>	>	>	>
Annual burden per capita	>	>	>	>	>	>
Total annual burden	>	>	>	>	>	>
Total savings (by subsidy beneficiaries)	<	<	<	<	<	<
Savings per capita (Annual)	<	<	<	<	<	<

Table B3

The comparison between the actual results from the original and the five sensitivity scenarios (as a basis for Table B2).

		Study model Equity	Study model Equality	SS1 Equity	SS1 Equality	SS2 Equity	SS2 Equality	SS3 Equity	SS3 Equality	SS4 Equity	SS4 Equality	SS5 Equity	SS5 Equality
Price increase (total population)	Million	38.42	40.40	38.69	42.17	38.25	39.04	38.57	40.04	38.43	40.91	47.97	35.67
Price decrease (total population)	Million	45.24	43.25	44.98	41.50	45.42	44.63	44.84	43.34	45.26	42.79	35.72	47.89
Price increase (urban pop %)	%	32%	42%	35%	46%	29%	38%	34%	42%	30%	42%	10%	42%
Price increase (rural pop %)	%	14%	6%	11%	4%	16%	9%	12%	6%	16%	7%	47%	0%
Price decrease (urban pop %)	%	10%	0%	12%	0%	9%	0%	8%	0%	12%	0%	32%	0%
Price decrease (rural pop %)	%	44%	52%	42%	50%	46%	53%	45%	52%	42%	51%	10%	57%
Annual burden per capita	US \$	\$ 23.3	47.77	23.11	47.23	23.41	47.35	26.76	56.32	20.07	39.41	9.73	65.51
Total annual burden	Million US \$	\$895.9	\$1930.4	\$894.4	\$1992.2	\$895.5	\$1848.7	\$1032.4	\$2255.4	\$771.3	\$1612.7	\$467.1	\$2336.9
Total savings	Million	\$895.2	\$747.1	\$894.4	\$765.2	\$895.5	\$725.7	\$1026.6	\$863.3	\$771.3	\$626.8	\$467.1	\$244.8
Savings per capita (Annual)	US \$	\$19.8	\$17.3	\$19.9	\$18.4	\$19.7	\$16.2	\$22.9	\$19.9	\$17.0	\$14.6	\$13.1	\$5.1

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